

Rank-Three Mapping-Class-Finite Representations in Genus Four: A Candidate Extension

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Abstract

We give a candidate extension to genus four of the rank-three finite-image argument developed in the companion genus-at-least-five candidate manuscript. For an irreducible rank-three system, the new work eliminates the dense rank-eight adjoint coefficient, the rank-five orthogonal coefficient, and the rank-six monomial coefficient on every finite-etale moduli cover. It then resolves the scalar equality case in formal propagation by comparing the adjoint Killing form with the standard rank-eight symplectic monodromy representation. The conclusion remains conditional on the fixed-part, Hodge, parabolic, canonical-Deligne, finite-cover, integrality, isomonodromy, and socle interfaces stated below. The argument is new and unrefereed, and is presented as a **CANDIDATE**, not as an established theorem.

Status and scope. Every result called a “candidate theorem” in this manuscript is conditional on the imported interface package in [section 2](#). The companion genus-at-least-five manuscript is itself an unrefereed candidate, not a published input. This paper is restricted to genus four and makes no lower-genus assertion.

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1 Statement and proof architecture

Let $\Sigma_{4,n}$ be an orientable genus-four surface with n punctures. The mapping class group $\text{Mod}_{4,n}$ acts on conjugacy classes of representations of $\pi_1(\Sigma_{4,n})$ by precomposition.

Candidate theorem 1.1 (Rank-three finite image in genus four). *Assume the published inputs and the companion-candidate interfaces recorded in [section 2](#). Let $n \geq 0$. If*

$$\rho : \pi_1(\Sigma_{4,n}) \longrightarrow \text{GL}_3(\mathbf{C})$$

has finite $\text{Mod}_{4,n}$ -orbit up to conjugacy, then ρ has finite image.

Landesman–Litt prove finite image under the strict inequality $r < \sqrt{g+1}$ [4, Theorem 1.2.1]. Their published theorem therefore does not directly treat $(g, r) = (4, 3)$. The companion manuscript proposes a rank-three theorem for $g \geq 5$ [6]; the present paper is a separate extension, and does not promote either candidate to established status.

The proof has five layers.

- (1) The fixed-part reduction leaves three coefficient ranks: the dense rank-eight adjoint coefficient, a rank-five orthogonal coefficient, and a rank-six monomial coefficient.
- (2) Exact no-high and high-Harder–Narasimhan ledgers are reconstructed in genus four. Universal-extension symmetry, normalizer cocycles, Killing and Spin parity, canonical-lattice pairing bounds, and low-degree section estimates eliminate every dense rank-eight branch.
- (3) The rank-five coefficient is eliminated by the first-super endpoint classification and its invariant symmetric form.

- (4) Exact monomial arithmetic leaves two no-high rows and one strict high-HN row for the rank-six coefficient. Symmetric universal extensions, parabolic stability, and the determinant-sign character eliminate them.
- (5) Projective finite-index descent and Artin devissage are unchanged. The only new propagation equality identifies a putative rank-eight adjoint coefficient with the standard genus-four cohomology system; their preserved bilinear forms are incompatible.

2 Published inputs and imported candidate interfaces

The dependency boundary is part of the statement. We use the following published results in the forms already isolated in the companion manuscript:

- Landesman–Litt’s projective globalization, fixed-part, Hodge, parabolic multiplication, rigidity, integrality, isomonodromy, and characteristic-socle interfaces [4, 5];
- the small-slope vector-bundle Clifford and generated-subbundle estimates of Brambila-Paz–Grzegorzczak–Newstead [1];
- the classification of holomorphic sections of universal curves by Earle–Kra [3], and Chen–Salter’s virtual non-splitting theorem in its stated range $g \geq 4$ [2].

The following items are imported from the unrefereed companion candidate [6]. They are hypotheses of Candidate theorem 1.1, not claims of published provenance:

- B1 the passage from an MCG-finite rank-three system to its actual multiplicity-free adjoint constituents on connected finite-etale moduli covers;
- B2 the canonical-Deligne and coparabolic convention, the fixed-part Hodge-line reduction, and the closed-descent interface;
- B3 the strict endpoint and first-super endpoint estimates, including the weight-one universal-extension classification;
- B4 the normalizer boundary-cocycle obstruction and the extension compatibility identities for bracket and invariant symmetric forms;
- B5 projective finite-index descent, direct Artin devissage, the exact trivial-coefficient gap, reducible unitary multiplicity control, and the remaining projective/linear integrality, unitary-conjugate, and characteristic-socle propagation steps.

For clarity, the imported projective-closure reduction is

projective closure	ad V
PGL_3 -dense	U_8
$\mathrm{PGL}_2 \simeq \mathrm{SO}_3$	$U_5 \oplus U_3$
monomial, component quotient S_3	$U_6 \oplus U_2$
monomial, component quotient C_3	$U_3^+ \oplus U_3^- \oplus \chi \oplus \chi^{-1}$.

Ranks at most three are already below the relevant fixed-part threshold in genus four. The new fixed-part work is therefore exactly the elimination of U_8 , U_6 , and U_5 carried out below.

Audit point 2.1 (Conditionality). Finite HN enumeration and fibrewise linear algebra do not certify the geometric interfaces in B1–B5. In particular, compilation of this paper or successful execution of finite verification scripts is not a proof of Candidate theorem 1.1.

3 Genus-four setup

Work on a very general smooth genus-four fibre C of the relevant finite cover. It is nonhyperelliptic. For a parabolically stable coefficient E_* of parabolic degree zero, let E denote the underlying canonical Deligne bundle and let $S = -\deg E$ be its total canonical parabolic weight. For a rank-zero Hodge vector, let $M \subset E$ be the saturated Hodge line, put $L = M^{-1}$ and $G = E/M$, and let $A \subset E^\vee \otimes \omega_C$ be the saturated evaluation kernel. In genus four set

$$q = \deg(M^\vee \otimes \omega_C) = 6 - \deg M.$$

When the high-HN part $N \subset A$ is nonzero, write $t = \operatorname{rk} N$ and $y = \deg N - S$. All saturations and brackets below are taken in the canonical parahoric lattice supplied by B2 and B4.

The next two sections give the exhaustive numerical ledgers for the dense rank-eight coefficient. Subsequent sections eliminate every listed row.

4 No-high reconstruction

For the rank-eight dense adjoint coefficient, the no-high inequalities give

$$S \leq \frac{16(8-g)}{9}, \quad \max(2g-1, 2g-2+S) \leq q \leq 14-S.$$

The quotient bundle has

$$(\operatorname{rk} G, \deg G, h^0(G)) = (7, q - (2g-2) - S, q - g + 1).$$

The exhaustive HN enumeration imposes strictly decreasing HN slopes. For $S = 0$ it uses positive-degree factors, as in the companion manuscript; for $S > 0$ it also permits degree-zero factors, and every factor has slope at most $2g-2$. The section bounds are applied on the very general, nonhyperelliptic genus-four fibre.

4.1 Numerical survivor table

S	Candidate q	Eliminated numerically	Survivors
0	7–14	12, 13	7, 8, 9, 10, 11, 14
2	8–12	none	8, 9, 10, 11, 12
3	9–11	11	9, 10
4	10	none	10

Thus 14 of 17 candidates survive. The first is

$$(S, q) = (0, 7), \quad (\operatorname{rk} G, \deg G, h^0(G)) = (7, 1, 4).$$

The exact BPGN small-slope inequality gives only $h^0(G) \leq 5$ for a semistable rank-seven, degree-one piece in genus four. Requiring four sections is therefore genuinely compatible with the cited bound. Section counting cannot remove this branch.

5 High-HN numerical frontier

Write $t = \operatorname{rk} N$ and $y = \deg N - S$. After the offset, Clifford, lower-slope, and q inequalities, the following necessary rows remain. They have not been geometrically realized as adjoint branches.

5.1 Numerical survivor table

t	y	q	Weight condition
1	3	9	$S \geq 4$
1	4	8–10	$S \geq 3$
1	5	7–11	$S \geq 2$
2	10	8	$S \geq 3$
2	11	7–9	$S \geq 2$
3	17	7	$S \geq 2$

The omitted top offset $y = (2g - 2)t$ gives a nonpositive-parabolic-degree dual quotient of the stable adjoint bundle and is impossible for that reason.

6 Genus-four branch analysis

6.1 The first no-high branch is eliminated

In the $(S, q) = (0, 7)$ branch, G has rank seven, degree one, and four sections. If $H \subsetneq G$ is a subbundle and $\tilde{H} \subset E$ its inverse image, stability of the degree-zero bundle E gives $\deg \tilde{H} < 0$, hence $\deg H \leq 0$. Thus G is stable. Its saturated evaluation image is generically generated of degree zero, so it is $W = \mathcal{O}_C^4$. The quotient is stable as well: a proper subbundle of $Q = G/W$ pulls back to a subbundle of G of degree at most zero. Therefore

$$L = M^{-1}, \quad W = \mathcal{O}_C^4 \subset G, \quad (\operatorname{rk} Q, \deg Q) = (3, 1).$$

Let $P \subset E$ be the inverse image of W . Since $H^0(P) = 0$ and $h^1(M) = 4$, the boundary of

$$0 \longrightarrow M \longrightarrow P \longrightarrow W \longrightarrow 0$$

is an isomorphism $e : H^0(W) \simeq H^1(M)$. Define the quotient-bracket component

$$f : M \otimes W \longrightarrow Q, \quad \phi : H^0(W) \longrightarrow H^0(Q \otimes L).$$

The map $b : \Lambda^2 P \rightarrow Q$ supplies information absent from pointwise Lie algebra. A Čech comparison of local splittings gives

$$f(v)_* e(u) = f(u)_* e(v) \quad \text{in } H^1(Q).$$

After Serre duality and $H^0(L\omega_C) \simeq W^\vee$, put $V_\phi = W / \ker \phi$ and $k = \dim V_\phi$. The induced map

$$\Theta : H^0(Q^\vee \omega_C) \longrightarrow H^0(L\omega_C) \otimes V_\phi^\vee$$

has image in $\operatorname{Sym}^2(V_\phi^\vee)$.

This symmetry forces $k \leq 1$. Here is the complete rank analysis. Let $I \subset Q$ be the saturated image of f , of rank r .

- If $r = 1$, stability gives $\deg I \leq 0$, so the line $I \otimes L$ has at most one section.
- If $r = 2$, then $-2 \leq \deg I \leq 0$. The quotient line $R = Q/I$ satisfies $h^0(R^\vee \omega_C) \leq 3$, while $h^0(Q^\vee \omega_C) \geq 8$. Thus $\text{rk } \Theta \geq 5$, excluding $k = 2$. If $k \geq 3$, three generically spanning sections of the rank-two bundle $I \otimes L$ have at least two independent wedges. Otherwise their Plücker kernel would be constant and give a global linear relation. This would be a pencil on a line of degree at most two, impossible on the nonhyperelliptic genus-four fibre.
- If $r = 3$ and $k = 3$, then Θ is injective, but its domain has dimension at least eight and $\dim \text{Sym}^2(\mathbb{C}^3) = 6$.
- If $r = 3$ and $k = 4$, write $H = Q \otimes L$, let $a : \mathcal{O}_C^4 \rightarrow H$, set $A = \ker a$, and let D be the Fitting divisor of its torsion cokernel, of length ℓ . Then $\deg A = \ell - 4$ and injectivity on constant sections gives $\ell \leq 3$. For the raw image $I_0 = \text{im}(a)$, local Smith form gives $\mathcal{O}_C(-D)I_0^\vee \subset H^\vee$. If $K_A = \ker(W^\vee \rightarrow H^0(A^\vee))$, the lower dimension bounds for $H^0(L\omega_C(-D))$ and K_A are respectively

$$(4, 3, 2, 1) \quad \text{and} \quad (1, 2, 2, 3)$$

for $\ell = 0, 1, 2, 3$. Choose nonproportional elements u and η in these spaces. Their product is a section $\alpha \in H^0(Q^\vee \omega_C)$ with $\Theta(\alpha) = u \otimes \eta$. A pure tensor is symmetric only when its factors are proportional, a contradiction.

Consequently $\text{rk } \phi \leq 1$. Put $W_0 = \ker \phi$, so $\dim W_0 \geq 3$. If $h^0(L) = 0$, the remaining self-block of β on W_0 vanishes. If $L \simeq \mathcal{O}_C(p)$, write that block as sA , where s is the unique section of L and $A : W_0 \rightarrow W$ is constant. Compatibility with the universal extension gives

$$W_0 \xrightarrow{A} W \xrightarrow{e} H^1(M) \xrightarrow{s*} H^1(\mathcal{O}_C),$$

with zero composite. The last map is an isomorphism by the exact sequence for $0 \rightarrow \mathcal{O}_C(-p) \rightarrow \mathcal{O}_C \rightarrow k(p) \rightarrow 0$, and e is an isomorphism, so $A = 0$. Thus in either case $\mathcal{O}_C^3 \subset \ker \beta$.

There is a second, shorter global obstruction once three kernel sections are available. Put $K = N_E(M)/M$. The normalizer acts on M through a character $\lambda : K \rightarrow \mathcal{O}_C$. For global $x, y \in H^0(K)$, local lifts satisfy

$$\partial[x, y] = \lambda(x)\partial y - \lambda(y)\partial x.$$

Here $\partial : H^0(G) \rightarrow H^1(M)$ is injective because $H^0(E) = 0$. Hence

$$[x, y] = \lambda(x)y - \lambda(y)x.$$

No generically independent three-dimensional subspace of an $\mathfrak{sl}_3$ quotient normalizer obeys this law. Regular orbit types have quotient normalizer dimension at most two. In the semisimple $(2, 1)$ case the quotient is $\mathfrak{sl}_2$ and $\lambda = 0$, so the law would make it abelian. For a minimal nilpotent $m = E_{12}$, use the quotient basis

$$D = \text{diag}(1/2, -1/2, 0), \quad H = \text{diag}(1/2, 1/2, -1), \quad E_{13}, \quad E_{32}.$$

The character is one on D and zero on the other three vectors. Its kernel is nonabelian, and its only two-dimensional abelian subalgebra is $\langle E_{13}, E_{32} \rangle$. If a three-space is not contained in that kernel, the displayed law requires eigenvalue one with multiplicity two on this two-dimensional

intersection. The actual eigenvalues are $(1 + 3c)/2$ and $(1 - 3c)/2$ for $D + cH + aE_{13} + bE_{32}$, and cannot both be one. This contradiction eliminates $(S, q) = (0, 7)$ without a determinant or relative-section endpoint.

6.2 The next no-high branch: $(S, q) = (0, 8)$

Now $(\text{rk } G, \deg G, h^0(G)) = (7, 2, 5)$ and $\deg L = 2$. Positive HN degrees give exactly

HN type	Section maximum
$(7, 2)$	5
$(1, 1) + (6, 1)$	$1 + 4$
$(2, 1) + (5, 1)$	$1 + 4$
$(3, 1) + (4, 1)$	$2 + 3$

The evaluation saturation is consequently either

$$\mathcal{O}_C^5 \subset G, \quad (\text{rk } Q, \deg Q) = (2, 2),$$

with Q semistable, or

$$J \oplus \mathcal{O}_C^4 \subset G, \quad \deg J = 1, \quad (\text{rk } R, \deg R) = (2, 1),$$

with R stable. In the second case the Čech argument must use the full evaluation injection $\mathcal{O}_C^5 \rightarrow G$, including the section of J ; using only $\mathcal{O}_C^4$ loses the boundary isomorphism.

We use two elementary linear-algebra lemmas to make the remaining argument explicit.

First, let $V \subset H^0(H)$ have dimension k , where H has rank two, and consider

$$\delta : \Lambda^2 V \longrightarrow H^0(\det H).$$

If $\ker \delta$ contains a decomposable vector $s \wedge t$, the saturation of s, t is a line subbundle with at least two sections. Otherwise $\mathbb{P}(\ker \delta)$ is disjoint from $\text{Gr}(2, V) \subset \mathbb{P}(\Lambda^2 V)$. Since the Grassmannian has dimension $2k - 4$, the projective dimension theorem gives

$$\dim \text{im}(\delta) \geq 2k - 3.$$

Second, a projective pencil of singular ternary quadrics has either a common linear factor or a common kernel point. Indeed, a nontrivial such pencil has a rank-two member: otherwise two independent rank-one members would have a rank-two sum. Such a member can be written xy . Write the other member as

$$ax^2 + by^2 + cz^2 + dxy + exz + fyz.$$

The determinant of the pencil vanishes identically only if $c = 0$, $ef = 0$, and $af^2 + be^2 = 0$. If $e = f = 0$, the pencil has the common kernel point $[0 : 0 : 1]$; if only e or only f vanishes, it has the common factor y or x , respectively.

Return first to the $\mathcal{O}_C^5$ evaluation case. Put $H = Q \otimes L$, let $V = \text{im } \phi$, and set $k = \dim V$. The same zero-weight condition gives $\det E \simeq \mathcal{O}_C$, hence $\det G \simeq L$ and $\det Q \simeq L$. The same symmetric- Θ argument gives the initial dichotomy: a rank-one saturated bracket image has $k \leq 2$, with equality only when $I \otimes L$ is a degree-three trigonal pencil; a rank-two image cannot have $k = 2$, so it has $k \geq 3$.

There is one degree-two issue before $\ker \phi$ becomes a normalizer kernel. If $W_0 = \ker \phi$ and L is ineffective, the remaining self-block vanishes and $W_0 \otimes \mathcal{O}_C \subset \ker \beta$. If $L \simeq \mathcal{O}_C(D)$ is effective, its unique section gives a self-block sA_0 . Extension compatibility only forces the image of A_0 into the one-dimensional kernel of

$$s_* : H^1(M) \longrightarrow H^1(\mathcal{O}_C),$$

so at least $\dim W_0 - 1$ normalizer sections are guaranteed. The full boundary is injective because $H^0(E) = 0$. Thus every case with $k \leq 1$ supplies at least three genuine normalizer sections and is eliminated by the boundary-cocycle obstruction. If $k = 2$, the ineffective- L case is eliminated as well, leaving only an apparently exceptional effective- L case. That exception is also impossible, but its lift defect must be retained.

Let $I \subset Q$ be the degree-one bracket image, put $A = I \otimes L$, and let $R_1 = Q/I$. Then A is a basepoint-free g_3^1 . Let

$$T = \text{im}(H^0(Q^\vee \omega_C) \longrightarrow H^0(I^\vee \omega_C)).$$

For nonzero $\alpha \in T$, symmetry says that the two-space $\alpha H^0(A)$ equals the fixed plane $V^\vee \subset H^0(L\omega_C)$. Two independent elements of T would therefore have a ratio preserving $H^0(A)$. Multiplication by that ratio has an eigenvector, so the ratio is constant on the nonvanishing locus of the eigenvector and hence globally. Thus $\dim T \leq 1$. Exactness and Riemann–Roch give

$$\dim T = (4 + h^0(Q)) - (2 + h^0(R_1)) = 2 + h^0(Q) - h^0(R_1).$$

Since $h^0(R_1) \leq 1$, one must have $R_1 \simeq \mathcal{O}_C(r)$ and $h^0(Q) = 0$. The inclusion $I \subset Q$ then gives $h^0(I) = 0$. Determinants give $I \otimes R_1 = L$, hence $I = L(-r)$ and $A = L^2(-r)$. The equality characterization in Clifford's theorem gives $h^0(L^2) \leq 2$ on a nonhyperelliptic genus-four curve. Since $h^0(A) = 2$, the point r would be a base point of $|L^2|$. But $h^0(I) = 0$ says that r is not in the divisor of the unique section of the effective degree-two line L , so the square of that section does not vanish at r . This contradiction eliminates the rank-one $k = 2$ case.

It remains to exclude full bracket image in Q . For $k = 3$, Θ is injective and

$$\dim \text{im } \Theta = 4 + h^0(Q), \quad h^0(Q) \leq 2.$$

On the dense open where $V \otimes \mathcal{O}_C \rightarrow H$ has rank two, let $\ell_x \in \mathbb{P}(V)$ be its kernel and let $u_x \in \mathbb{P}(V)$ be the evaluation vector of the fixed inclusion $V^\vee \subset H^0(L\omega_C)$. If M_α is the symmetric matrix $\Theta(\alpha)$, evaluating on ℓ_x gives

$$u_x^t M_\alpha \ell_x = 0.$$

Consequently the singular conic $q_x = \text{sym}(u_x \otimes \ell_x)$ lies in $(\text{im } \Theta)^\perp \subset \text{Sym}^2(V)$. If $h^0(Q) = 2$ this orthogonal space is zero. If $h^0(Q) = 1$ it is a fixed projective point, and unique factorization of the fixed reducible conic makes either $[u_x]$ or $[\ell_x]$ constant, contradicting independence of the dual sections or of V . If $h^0(Q) = 0$, the orthogonal space is a projective pencil. The map $x \mapsto [q_x]$ cannot be constant by the same factorization argument, so the whole pencil is singular. The second lemma gives either a common factor, again making one of $[u_x], [\ell_x]$ constant, or a common kernel point. In the latter case all u_x lie in a fixed two-plane, so a nonzero member of $V^\vee$ vanishes identically. Every alternative is impossible.

For $k = 4$ or 5 , apply the wedge lemma to H . Here H is semistable of rank two and degree six, and $\det H = L^3$ has at most four sections. If $k = 5$, the lemma forces a line subbundle $A_1 \subset H$ with

at least two sections. Semistability and nonhyperellipticity force $\deg A_1 = 3$ and $h^0(A_1) = 2$; the degree-three quotient has at most two sections, contradicting $k = 5$.

If $k = 4$, the same lemma forces a line subbundle, and equality in the section bounds forces an exact sequence

$$0 \longrightarrow A_1 \longrightarrow H \longrightarrow A_2 \longrightarrow 0$$

in which A_1, A_2 are basepoint-free g_3^1 's and every section of A_2 lifts. A base point in either pencil would leave a degree-two pencil and make C hyperelliptic. In particular $V = H^0(H) = V_1 \oplus V_2$, with $V_1 = H^0(A_1)$ and V_2 a lift of $H^0(A_2)$. Twisting by M gives $I = A_1 \otimes M \subset Q$ and $R_2 = A_2 \otimes M = Q/I$. For $\alpha \in H^0(R_2^\vee \omega_C) \subset H^0(Q^\vee \omega_C)$, the V_1 columns of $\Theta(\alpha)$ vanish; symmetry makes its V_1 rows vanish as well. Hence $\alpha H^0(A_2)$ is the fixed two-plane $V_2^\vee$. But

$$h^0(R_2^\vee \omega_C) = 2 + h^0(R_2) \geq 2,$$

so two independent choices of α give the same product pencil. Their ratio preserves $H^0(A_2)$ and is constant by the eigenvector argument, a contradiction. Thus every $\mathcal{O}_C^5$ evaluation case is eliminated.

For the $J \oplus \mathcal{O}_C^4$ evaluation, the full map $\mathcal{O}_C^5 \rightarrow G$ is not saturated at the zero of the section of J . It is enough for the Čech symmetry, but bracket with M is not known to preserve that raw evaluation lattice. Its self-block may acquire a pole in a raw trivialization. The full-rank residual cases can nevertheless be excluded before confronting that lattice issue. Write the stable residual bundle as R_0 and put $H_0 = R_0 \otimes L$, a stable rank-two bundle of degree five. If H_0 had four sections, the wedge lemma and $h^0(\det H_0) \leq 3$ would produce a line subbundle with two sections; stability gives degree at most two, contradicting nonhyperelliptic gonality. Thus $h^0(H_0) \leq 3$.

A full bracket image cannot have coefficient rank two, because injectivity of Θ gives

$$\dim H^0(R_0^\vee \omega_C) = 5 + h^0(R_0) > 3 = \dim \text{Sym}^2(\mathbb{C}^2).$$

It must therefore have $k = 3$. The same q_x construction now has $\dim \text{im } \Theta = 5$ or 6 : indeed, stability gives $h^0(R_0) \leq 1$. To see this, two sections with zero wedge would lie in a degree-zero line with two sections, while a nonzero wedge vanishes at the unique point of $\det R_0$ and a linear combination would generate a degree-at-least-one line subbundle. Both alternatives contradict stability. Thus the orthogonal of $\text{im } \Theta$ in $\text{Sym}^2(\mathbb{C}^3)$ is projectively a point or empty. The empty case is immediate; in the point case unique factorization again forces an evaluation vector or a kernel line to be constant. Hence every full-rank residual image is impossible. A rank-one residual image has $k \leq 1$: its saturated line in R_0 has degree at most zero, so after twisting by L it has degree at most two and at most one section. It remains to convert the resulting large kernel into genuine normalizer sections without pretending that the raw evaluation lattice is saturated.

Let

$$V_0 = H^0(G), \quad F = \text{im}(V_0 \otimes \mathcal{O}_C \rightarrow G), \quad W = J \oplus \mathcal{O}_C^4.$$

The five sections are generically independent, so $F \simeq \mathcal{O}_C^5$, and the degree-one saturation defect gives $W/F \simeq k(p)$. Stability gives $H^0(E) = 0$; since $h^1(M) = 5$, the extension boundary is an isomorphism

$$e : V_0 \xrightarrow{\sim} H^1(M).$$

Write $\phi : V_0 \rightarrow H^0(R_0 \otimes L)$ for the residual quotient-bracket map. The Čech symmetry is valid for the actual raw sections: choosing local lifts in E and projecting their bracket to R_0 gives

$$\phi(v)_*e(u) = \phi(u)_*e(v) \quad \text{in } H^1(R_0).$$

Suppose first that $\dim \operatorname{im} \phi = 1$. Write $\phi(v) = c(v)r$ with $0 \neq r : M \rightarrow R_0$. Taking $u \in \ker c$ and v with $c(v) = 1$ shows that $r_*e(u) = 0$, hence

$$\dim \ker(H^1(M) \xrightarrow{r_*} H^1(R_0)) \geq 4.$$

Let $I \subset R_0$ be the saturation of $r(M)$ and put $d = \operatorname{length}(I/r(M))$. Stability of the rank-two, degree-one bundle R_0 gives $\deg I \leq 0$, so $0 \leq d \leq 2$. For $Q_r = R_0/r(M)$ there is an exact sequence

$$0 \longrightarrow I/r(M) \longrightarrow Q_r \longrightarrow R_0/I \longrightarrow 0, \quad \deg(R_0/I) = 3 - d.$$

On the nonhyperelliptic genus-four test fibre, $h^0(R_0/I) \leq 2, 1, 1$ for $d = 0, 1, 2$, respectively. Thus $h^0(Q_r) \leq 2, 2, 3$. The long exact sequence of $0 \rightarrow M \rightarrow R_0 \rightarrow Q_r \rightarrow 0$ gives

$$\dim \ker r_* = h^0(Q_r) - h^0(R_0) \leq h^0(Q_r) \leq 3,$$

contradicting the preceding lower bound. Therefore $\phi = 0$.

Now $\beta(V_0) \subset H^0(W \otimes L)$. Compose with the principal-part map to $H^0((W/F) \otimes L)$ and let V_1 be its kernel. The target is one-dimensional, so $\dim V_1 \geq 4$, and

$$\beta(V_1) \subset H^0(F \otimes L) = V_0 \otimes H^0(L).$$

If $h^0(L) = 0$, all of V_1 consists of genuine normalizer sections. If $h^0(L) = 1$, write its generator as s and $\beta|_{V_1} = sA$ for a linear map $A : V_1 \rightarrow V_0$. The global bracket map $M \otimes G \rightarrow E$ lifts $sA(x) : M \rightarrow G$ for every $x \in V_1$. Hence the pullback of the extension along multiplication by s splits, or $s_*e(Ax) = 0$.

The zero divisor of s has length two, and $0 \rightarrow M \xrightarrow{s} \mathcal{O}_C \rightarrow \mathcal{O}_D \rightarrow 0$ gives $\dim \ker(s_* : H^1(M) \rightarrow H^1(\mathcal{O}_C)) = 1$. Since e is an isomorphism, $\operatorname{rk} A \leq 1$ and therefore $\dim \ker A \geq 3$. These are three generically independent global normalizer sections because $F \simeq V_0 \otimes \mathcal{O}_C$. The normalizer boundary-cocycle obstruction excludes them. Thus the entire nonsaturated evaluation locus, and hence all of $(S, q) = (0, 8)$, is impossible.

6.3 The zero-weight $q = 9$ branch and its closure

For $(S, q) = (0, 9)$ one has

$$\deg M = -3, \quad \deg L = 3, \quad (\operatorname{rk} G, \deg G, h^0(G)) = (7, 3, 6).$$

Stability gives $H^0(E) = 0$, so the full boundary is an isomorphism

$$e : H^0(G) \xrightarrow{\sim} H^1(M)$$

between six-dimensional spaces. The exact HN equality types allowed by the small-slope bounds are

$$(7, 3), \quad (1, 1) + (6, 2), \quad (2, 2) + (5, 1).$$

The saturation W of the six-section evaluation has rank six. Stability of E bounds its degree by two, and equality in the section bounds gives exactly

$$\mathcal{O}_C^6, \quad J_1 \oplus \mathcal{O}_C^5, \quad A_2 \oplus \mathcal{O}_C^5, \quad B_{2,2} \oplus \mathcal{O}_C^4.$$

Here J_1 is a degree-one line with one section, A_2 is a degree-two line with one section, and $B_{2,2}$ is a semistable rank-two, degree-two bundle with two sections. If $d = \deg W \in \{0, 1, 2\}$ and $F = \text{im}(H^0(G) \otimes \mathcal{O}_C \rightarrow G)$, then

$$F \simeq \mathcal{O}_C^6, \quad \text{length}(W/F) = d, \quad R := G/W \simeq L(-D), \quad \deg D = d.$$

Let

$$\phi : H^0(G) \longrightarrow H^0(R \otimes L), \quad U = \text{im } \phi, \quad k = \dim U$$

be the residual bracket coefficient. If $\phi \neq 0$, Čech symmetry gives an injective map

$$\Theta : H^0(R^{-1} \otimes \omega_C) \longrightarrow \text{Sym}^2(U^\vee).$$

Every nonzero form in its image is nondegenerate. Indeed, choose $0 \neq u \in U$ and a lift $v \in H^0(G)$ with $\phi(v) = u$. If u were a null vector, then for every $w \in H^0(G)$ the Čech symmetry would give

$$0 = \langle e(v), \alpha\phi(w) \rangle = \langle e(w), \alpha u \rangle.$$

The boundary isomorphism and Serre duality would force $\alpha u = 0$, which is impossible for two nonzero sections on an integral curve. A complex pencil of symmetric forms always contains a singular member, so

$$\phi \neq 0 \implies h^0(R^{-1} \otimes \omega_C) \leq 1.$$

Riemann–Roch and $R = L(-D)$ give the exact dimension

$$h^0(R^{-1} \otimes \omega_C) = d + h^0(R).$$

Consequently $d = 2$ forces $k = 0$; $d = 1$ forces $k = 0$ when R is effective; and $d = 0$ forces $k = 0$ when L is trigonal.

The remaining issue is no longer the residual Petri map. After taking $\ker \phi$ and killing the d principal parts of W/F , there is a subspace $V_1 \subset H^0(G)$ of dimension at least $6 - k - d$ on which the self-block is

$$V_1 \longrightarrow H^0(G) \otimes H^0(L).$$

Write $\ell = h^0(L)$. On the nonhyperelliptic genus-four test fibre, $\ell \in \{0, 1, 2\}$. Extension compatibility places the self-block in the kernel of

$$\delta_L : H^1(M) \otimes H^0(L) \longrightarrow H^1(\mathcal{O}_C), \quad a \otimes s \longmapsto s_* a.$$

For $\ell = 0, 1, 2$, the kernel dimensions are respectively

$$\kappa_\ell = 0, 2, 8.$$

For $\ell > 0$ this follows because multiplication by any nonzero section of L surjects $H^1(M) \rightarrow H^1(\mathcal{O}_C)$; the domain of δ_L has dimension 6ℓ and the target has dimension four. The guaranteed normalizer rank is therefore only

$$N = \max\{0, 6 - k - d - \kappa_\ell\}.$$

This formula closes some subloci, but not the branch:

Evaluation data	Residual rank	Guaranteed N	Status from the extension count alone
$d = 2, \ell = 0$	$k = 0$	4	closed
$d = 2, \ell = 1$	$k = 0$	2	not closed at this stage
$d = 2, \ell = 2$	$k = 0$	0	not closed at this stage
$d = 1, R$ effective, $\ell = 1$	$k = 0$	3	closed
$d = 1, R$ effective, $\ell = 2$	$k = 0$	0	not closed at this stage
$d = 0, \ell = 2$	$k = 0$	0	not closed at this stage

There are also possible escapes with nonzero residual coefficient rank: $d = 0, \ell = 0, k = 4$ when $h^0(L^2) = 4$, equivalently $L^2 \simeq \omega_C$; $d = 0, \ell = 1, k \geq 2$; and, when R is ineffective, $d = 1, \ell = 0, k = 3$ when $h^0(R \otimes L) = 3$, or $d = 1, \ell = 1, k \geq 1$. The case $d = 1, R$ ineffective, $\ell = 2$ cannot occur because a degree-three pencil on the nonhyperelliptic fibre is basepoint-free. In each surviving case the current lower bound can fall below three.

The global Killing pairing supplies the missing constraint. Let $Q_2 = \text{tr}(m^2) \in H^0(L^2)$. Pairing the inverse image $P \subset E$ with M by the global Killing form gives

$$b : P \longrightarrow M^* = L, \quad b|_M = Q_2.$$

Thus the pushout of the universal extension $0 \rightarrow M \rightarrow P \rightarrow \mathcal{O}_C^6 \rightarrow 0$ along $Q_2 : M \rightarrow L$ splits. Since its boundary $e : \mathbb{C}^6 \rightarrow H^1(M)$ is an isomorphism, multiplication

$$(Q_2)_* : H^1(M) \longrightarrow H^1(L)$$

must vanish. If $Q_2 \neq 0$, its Serre-dual map is multiplication by Q_2 from $H^0(\omega_C \otimes L^{-1})$ to $H^0(\omega_C \otimes L)$; this is injective on the integral curve and has nonzero source, of dimension $h^0(L) \geq 1$. Hence $(Q_2)_*$ cannot vanish. Therefore every effective- L realization, equivalently every sublocus with $h^0(L) > 0$, must satisfy

$$Q_2 = \text{tr}(m^2) = 0.$$

In the unpunctured no-high branch, Killing gives more than isotropy. Since $Q_2 = 0$, pairing with M descends to

$$\bar{b} : G = E/M \longrightarrow L.$$

For $x, y \in H^0(G)$, choose local lifts with boundary cocycles representing $e(x), e(y) \in H^1(M)$. The overlap formula for their Killing pairing gives

$$\bar{b}(y)_* e(x) + \bar{b}(x)_* e(y) = 0 \quad \text{in } H^1(\mathcal{O}_C).$$

If the induced map $H^0(G) \rightarrow H^0(L)$ were nonzero, choose x whose image is a nonzero section s . Its kernel has dimension at least $6 - h^0(L) \geq 4$, and the displayed identity would carry that kernel injectively under e into

$$\ker(s_* : H^1(M) \longrightarrow H^1(\mathcal{O}_C)),$$

which has dimension two. This is impossible. Hence $\bar{b}$ kills the raw evaluation $F \simeq \mathcal{O}_C^6$. If $P \subset E$ is its inverse image, then $P \subset M^\perp$; both bundles have rank seven and degree -3 , so

$$P = M^\perp, \quad F = M^\perp/M.$$

In particular F is saturated. Thus every effective- L case with $d = 1$ or $d = 2$ is impossible. Moreover $[M, M^\perp] \subset M^\perp$, so the residual coefficient map vanishes automatically when $d = 0$. If $h^0(L) = 1$, the remaining self-block is sA and compatibility puts $e(\text{im } A)$ in the two-dimensional kernel of s_* . Hence $\text{rk } A \leq 2$ and at least four constant normalizer sections remain, a contradiction. The last effective case left by these extension counts is

$$d = 0, \quad h^0(L) = 2, \quad P = M^\perp.$$

Here Killing descends to a nondegenerate constant symmetric form on $F \simeq \mathcal{O}_C^6$, and the self-block is a pencil skew for that form.

The actual inner PGL_3 structure now gives a topological contradiction. The adjoint homomorphism

$$\text{PGL}_3 \longrightarrow \text{SO}_8$$

lifts to Spin_8 : lift first from the simply connected group SL_3 , and observe that its central μ_3 must map trivially to the μ_2 kernel of $\text{Spin}_8 \rightarrow \text{SO}_8$. Hence the orthogonal obstruction class of the adjoint bundle is $w_2(E) = 0$. On the other hand, a smooth orthogonal splitting along the isotropic line gives

$$E \simeq H(M) \perp F, \quad w_2(E) = c_1(M) + w_2(F) \pmod{2}.$$

The form on the trivial bundle $F \simeq \mathcal{O}_C^6$ is a constant nondegenerate matrix, so $w_2(F) = 0$, while $\deg M = -3$ is odd. This gives $w_2(E) = 1$, a contradiction. Here w_2 is the obstruction class of the complex orthogonal bundle, not the Stiefel–Whitney class of its underlying realification. Therefore **every effective- L no-high $q = 9$ sublocus is impossible.**

The punctured canonical lattices admit an even sharper elementary obstruction. At a marked DVR their standard basis consists of two Cartan generators and root generators

$$z^{\epsilon_{ij}} E_{ij}, \quad \epsilon_{ij} \in \{0, 1\}.$$

For an opposite root pair, $\epsilon_{ij} + \epsilon_{ji}$ is zero inside a Levi block and one across distinct flag blocks. Let a saturated, hence primitive, isotropic generator m have pairing ideal $\text{tr}(mE) = z^{d_p} R$. If a unit coefficient of m occurs in the Cartan part, the unimodular Cartan Gram matrix gives $d_p = 0$. If it occurs on a root generator, pairing with the unique opposite root generator isolates that coefficient and gives $d_p \leq 1$. Thus

$$d_p \leq 1$$

at every nonscalar marking, with no possible cancellation; at scalar or ordinary points $d_p = 0$.

Let N_{ns} be the number of nonscalar markings and let s_p be the adjoint determinant contribution, two for type $2 + 1$ and three for a full flag. If $D_b = \sum_p d_p p$, then

$$\deg D_b \leq N_{\text{ns}} \leq S/2, \quad S = \sum_p s_p.$$

On the other hand,

$$\text{im}(E \longrightarrow M^*) = L(-D_b), \quad \deg M^\perp = \deg D_b - S - 3.$$

The symmetric Čech identity gives $P \subset M^\perp$, while $\deg P = -3$, and therefore $\deg D_b \geq S$. This contradicts the displayed upper bound whenever $S > 0$. Hence **every effective punctured high-HN $q = 9$ row is impossible**. This argument uses the canonical periodic-chain root exponents; it is not asserted for an arbitrary commensurable lattice.

The ineffective- L subloci close by a different two-normalizer argument. Put $\ell = h^0(L) = 0$. The extension count already closes $d = 2$. For $d = 1$, the only possible escape has $k = 3$: the three-dimensional space $\ker \phi$ loses at most one dimension to the length-one principal-part map. Its remaining two-space has bracket in $H^0(F \otimes L) = H^0(L)^6 = 0$. For $d = 0$, the only escape has $k = 4$ and the two-dimensional space $\ker \phi$ has the same property. Thus in either case there are two generically independent global sections

$$\mathcal{O}_C^2 \longrightarrow K := \ker \beta.$$

The two maximal escapes have the exact line-bundle forms

$$(d, k) = (0, 4) : \quad L^2 \simeq \omega_C,$$

and

$$(d, k) = (1, 3) : \quad L^2 \simeq \omega_C(D - p)$$

for an effective divisor D of degree one and a point p . Lower coefficient rank already leaves at least three normalizer sections.

Let $Q_2 = \text{tr}(m^2) \in H^0(L^2)$. If $Q_2 = 0$, Killing pairing descends on the inverse image P of the raw $F \simeq \mathcal{O}_C^6$ to a map $F \rightarrow L$. This map is zero because $h^0(L) = 0$, so $P \subset M^\perp$. Equal rank and degree give $P = M^\perp$. For $d = 1$ this already contradicts the nonsaturation of F ; for $d = 0$ it gives $M^\perp/M \simeq \mathcal{O}_C^6$ and the Spin-parity contradiction above.

It remains to take $Q_2 \neq 0$. For a quotient-normalizer element x , write $[x, m] = \lambda(x)m$. Killing invariance gives

$$\lambda(x)Q_2 = \text{tr}([x, m]m) = 0,$$

so the grading character λ vanishes. The injective-boundary Čech law then makes the two global normalizer sections commute. After extending the generic field if necessary, if m is regular, its centralizer has dimension two and $Z(m)/\mathbb{C}m$ has dimension one, so it cannot contain them. If m is nonregular and $Q_2 \neq 0$, its minimal polynomial has degree at most two and tracelessness forces the semisimple eigenvalue pattern $(a, a, -2a)$ with $a \neq 0$. In that case

$$Z(m)/\mathbb{C}m \simeq \mathfrak{sl}_2,$$

which has no two-dimensional abelian subalgebra. This is the final contradiction. Therefore **the entire unpunctured no-high branch $(S, q) = (0, 9)$ is impossible**.

6.4 The first punctured branch: $(S, q) = (2, 8)$

Here the branch data are

$$\deg M = -2, \quad \deg L = 2, \quad (\text{rk } G, \deg G, h^0(G)) = (7, 0, 5), \quad h^1(M) = 5.$$

The no-high bundle A has rank seven, degree 42, and $\mu_{\max}(A) \leq 6 = \mu(A)$. Hence A is semistable, and $G = A^\vee \otimes \omega_C$ is semistable of degree zero. The saturated evaluation image is

generically generated, so it has nonnegative degree; semistability of G gives the opposite inequality. It is therefore a degree-zero trivial bundle. Since all five global sections of G factor through it, its rank is at least five; an evaluation image generated by five sections has rank at most five. Thus its rank is exactly five:

$$W = \mathcal{O}_C^5 \subset G.$$

The quotient $Q = G/W$ is semistable of rank two and degree zero. Parabolic stability gives $H^0(E) = 0$, so the boundary of the inverse-image extension is an isomorphism

$$e : H^0(W) \xrightarrow{\sim} H^1(M).$$

Put $H = Q \otimes L$, a semistable rank-two bundle of degree four. On the nonhyperelliptic test fibre, Riemann–Roch gives

$$h^0(\det H) = 1 + h^0(\omega_C \otimes \det H^{-1}) \leq 2,$$

because the residual line has degree two. Three independent sections of H would give a nonzero kernel for

$$\Lambda^2 \mathbb{C}^3 \longrightarrow H^0(\det H).$$

Every nonzero bivector in three variables is decomposable, so two independent sections would generate a line subbundle of degree at most two. Such a line cannot have two sections on a nonhyperelliptic curve. Therefore

$$h^0(Q \otimes L) \leq 2.$$

Let $\phi : H^0(W) \rightarrow H^0(Q \otimes L)$ be the quotient-bracket coefficient map and put $k = \dim \operatorname{im} \phi$. If $k = 2$, the two image sections generically span $Q \otimes L$; otherwise they again generate a forbidden degree-at-most-two pencil. The Čech symmetry then makes the Petri map injective with image in the symmetric square:

$$H^0(Q^\vee \otimes \omega_C) \hookrightarrow \operatorname{Sym}^2(\mathbb{C}^2)^\vee.$$

Riemann–Roch gives $h^0(Q^\vee \otimes \omega_C) = 6 + h^0(Q) \geq 6$, while the target has dimension three. Thus $k \leq 1$.

Now $W_0 = \ker \phi$ has dimension at least four. If L is ineffective, the remaining self-block vanishes. If $L \simeq \mathcal{O}_C(D)$, write the block as sA . Extension compatibility gives $s_*eA = 0$, and the exact sequence

$$0 \longrightarrow M \xrightarrow{s} \mathcal{O}_C \longrightarrow \mathcal{O}_D \longrightarrow 0$$

shows that $\ker(s_* : H^1(M) \rightarrow H^1(\mathcal{O}_C))$ is one-dimensional. Since e is an isomorphism, $\operatorname{rk} A \leq 1$. At least three generically independent constant sections therefore lie in $\ker \beta$, and the normalizer boundary-cocycle obstruction eliminates the branch. No Pfaffian effectivity or relative-section endpoint is used.

6.5 The punctured no-high $q = 9$ rows are eliminated

The two punctured no-high numerical rows at coefficient rank $q = 9$ are $(S, q) = (2, 9), (3, 9)$.

For $(S, q) = (2, 9)$ one has

$$(\mathrm{rk} G, \deg G, h^0(G)) = (7, 1, 6), \quad \deg M = -3, \quad h^1(M) = 6.$$

Let W be the saturation of the six-section evaluation and F its raw image. The no-high slopes of G are nonnegative. Since G/W also has nonnegative degree, one has $0 \leq \deg W \leq 1$. A rank-at-most-five bundle of degree at most one that is generically generated cannot carry six independent sections: after choosing its rank many independent sections it is either $\mathcal{O}_C^r$ or $\mathcal{O}_C(p) \oplus \mathcal{O}_C^{r-1}$, and in either case has at most r sections. Hence W has rank six. The evaluation $H^0(G) \otimes \mathcal{O}_C \rightarrow G$ is therefore generically injective, so $F \simeq \mathcal{O}_C^6$, with the exact alternatives

$$W = F = \mathcal{O}_C^6 \quad \text{or} \quad W = \mathcal{O}_C(p) \oplus \mathcal{O}_C^5, \quad W/F \simeq k(p).$$

Write $\delta = \deg W \in \{0, 1\}$ and $R = G/W$, so $\deg R = 1 - \delta$. If the residual bracket coefficient $\phi : \mathbb{C}^6 \rightarrow H^0(R \otimes L)$ were nonzero, Čech symmetry would inject

$$H^0(R^{-1} \otimes \omega_C) \longrightarrow \mathrm{Sym}^2(\mathrm{im} \phi)^\vee$$

with every nonzero form in the image nondegenerate. But Riemann–Roch gives

$$h^0(R^{-1} \otimes \omega_C) = 2 + \delta + h^0(R) \geq 2,$$

If $\dim \mathrm{im} \phi = 1$, the target of the displayed injection has dimension one, already contradicting the source dimension. If $\dim \mathrm{im} \phi \geq 2$, any two-dimensional subspace of the image is a complex pencil of symmetric forms and contains a singular member. Thus $\phi = 0$.

If L is ineffective, killing the at-most-one principal part leaves at least five raw sections, and their self-block lies in $H^0(F \otimes L) = 0$. The normalizer boundary law gives a contradiction. If L is effective, the quadratic Killing pushout forces $Q_2 = \mathrm{tr}(m^2) = 0$. The symmetric Killing Čech identity then gives $P \subset M^\perp$. Since $S = 2$ consists of one type-2 + 1 marking, the canonical local basis gives $\deg D_b \leq 1$, while the global degree comparison gives $\deg D_b \geq S = 2$. This is impossible.

For $(S, q) = (3, 9)$ the data are

$$(\mathrm{rk} G, \deg G, h^0(G)) = (7, 0, 6), \quad \deg M = -3, \quad h^1(M) = 6.$$

Here the no-high inequalities make G semistable of degree zero, so its saturated evaluation is

$$W = \mathcal{O}_C^6,$$

with degree-zero quotient line Q . A nonzero residual coefficient would again produce a subspace of everywhere nondegenerate symmetric forms, now of dimension

$$h^0(Q^{-1} \otimes \omega_C) = 3 + h^0(Q) \geq 3,$$

which is impossible. Thus the residual coefficient vanishes. If L is ineffective, all six raw sections lie in the quotient normalizer. If L is effective, the same Killing argument gives $P \subset M^\perp$. The equation $2a + 3b = 3$ forces exactly one full-flag marking, so the canonical local bound is $\deg D_b \leq 1$, while the global inclusion requires $\deg D_b \geq 3$. This is again impossible.

Therefore **all no-high genus-four rows at $q = 9$ are impossible**.

6.6 The no-high $q = 10$ layer is eliminated

The four rows at $q = 10$ have the common data

$$\deg M = -4, \quad \deg L = 4, \quad h^1(M) = h^0(G) = 7, \quad \deg G = 4 - S.$$

For $S = 0$, the unique HN type attaining the required seven-section upper bound is

$$(\mathrm{rk}, \deg) = (2, 3) + (5, 1),$$

with section contributions three and four. Let H be the semistable rank-two, degree-three first factor. Equality would give $h^0(H) = 3$. The wedge map

$$\Lambda^2 H^0(H) \longrightarrow H^0(\det H)$$

is injective: a nonzero kernel vector is decomposable because $\dim H^0(H) = 3$, and its two sections would generate a line subbundle with at least two sections. Such a line has degree at least three on the nonhyperelliptic test fibre, contradicting semistability of H of slope $3/2$. But $\det H$ has degree three and at most two sections, so the three-dimensional exterior square cannot inject. The row $(0, 10)$ is impossible.

Now let $S = 2, 3, 4$. The exact equality HN types are respectively

$$(1, 2) + (6, 0) \text{ or } (2, 2) + (5, 0), \quad (1, 1) + (6, 0), \quad (7, 0).$$

Equality forces every degree-zero factor to be trivial and the positive factor to have full-rank evaluation. Hence the seven sections have generically full-rank raw image

$$F \simeq \mathcal{O}_C^7 \subset G, \quad d := \mathrm{length}(G/F) = 4 - S = 2, 1, 0.$$

If $P \subset E$ is the inverse image of F , parabolic stability gives $H^0(P) = 0$, so the extension boundary is an isomorphism

$$e : \mathbb{C}^7 \xrightarrow{\sim} H^1(M).$$

Riemann–Roch and nonhyperellipticity give $h^0(L) \in \{1, 2\}$. Suppose first that $h^0(L) = 1$, with generator s . Killing the d principal parts leaves a subspace V_1 of dimension at least $7 - d$ on which the self-block has the form sA . Extension compatibility gives

$$s_* eA = 0,$$

and the degree-four divisor sequence gives

$$\dim \ker(s_* : H^1(M) \longrightarrow H^1(\mathcal{O}_C)) = 3.$$

Thus at least $7 - d - 3 = 4 - d$ constant quotient-normalizer directions remain. For $S = 3, 4$ this is at least three, and the normalizer boundary-cocycle obstruction closes the rows.

For $S = 2$, the count leaves two directions. Put $Q_2 = \mathrm{tr}(m^2)$. If $Q_2 = 0$, the Killing pairing descends to a map $G \rightarrow L$. If its restriction to the seven raw sections were nonzero, the symmetric Killing Čech identity would inject a subspace of dimension at least six under e into the three-dimensional kernel of multiplication by s . Hence that restriction vanishes. Since F has full generic rank in G , the Hodge line would then be orthogonal to the whole generic fibre of E , contradicting nondegeneracy of the $\mathfrak{sl}_3$ Killing form.

If instead $Q_2 \neq 0$, Killing invariance forces the grading character to vanish on the two surviving quotient-normalizer sections, and the boundary law makes them commute. A regular element has $\dim Z(m)/\mathbb{C}m = 1$. The only nonregular traceless 3×3 orbit with $Q_2 \neq 0$ is semisimple of type $(2, 1)$, where $Z(m)/\mathbb{C}m \simeq \mathfrak{sl}_2$, which contains no abelian two-plane. This closes the last one-section case.

Finally suppose $h^0(L) = 2$. Riemann–Roch gives $h^0(\omega_C \otimes L^{-1}) = 1$. The Killing map splits the pushout of the universal extension along $Q_2 : M \rightarrow L$, so $(Q_2)_*e = 0$. Since e is surjective, $(Q_2)_* = 0$ on $H^1(M)$. If Q_2 were nonzero, its Serre-dual multiplication map from the one-dimensional space $H^0(\omega_C \otimes L^{-1})$ would be injective. Therefore $Q_2 = 0$, and the preceding full-rank Killing contradiction applies. Consequently **every no-high genus-four row at $q = 10$ is impossible**.

6.7 A degree-four factor lemma

We use the following sharpening repeatedly. If H is semistable of degree four and rank $r = 1, 2, 3$ on the nonhyperelliptic genus-four test fibre, then

$$h^0(H) \leq r + 1.$$

For $r = 1$, Riemann–Roch reduces the assertion to the fact that a degree-two line has at most one section. For $r = 2$, suppose $h^0(H) = 4$ and put $V = H^0(H)$. Since $h^0(\det H) \leq 2$, the kernel of

$$\Lambda^2 V \longrightarrow H^0(\det H)$$

has dimension at least four. Its projectivization has dimension at least three and must meet the four-dimensional Grassmannian $\mathrm{Gr}(2, 4) \subset \mathbb{P}^5$. A decomposable kernel vector would give a line subbundle with two sections and hence degree at least three, contradicting semistability at slope two.

For $r = 3$, suppose $h^0(H) = 5$. Every saturated rank-two subsheaf $K \subset H$ has degree at most two and cannot have three sections: the exterior square of three such sections maps to the at-most-one-dimensional space $H^0(\det K)$, so a decomposable kernel vector would give a two-section line subbundle, impossible under semistability of H . On the other hand, the kernel of

$$\Lambda^3 H^0(H) \longrightarrow H^0(\det H)$$

has dimension at least eight. Its projectivization is at least $\mathbb{P}^7$ and must meet $\mathrm{Gr}(3, 5) \subset \mathbb{P}^9$, of dimension six. The resulting three sections lie in a saturated subsheaf of rank at most two, which was just excluded. This proves the lemma.

6.8 The no-high $q = 11$ layer is eliminated

For $(S, q) = (0, 11)$ one has

$$(\mathrm{rk} G, \deg G, h^0(G)) = (7, 5, 8).$$

The only HN types whose numerical bounds reach eight sections are

$$(2, 4) + (5, 1), \quad (3, 4) + (4, 1),$$

with respective section bounds $4 + 4$ and $5 + 3$. The degree-four factor lemma improves the leading bounds to three and four, so both totals are at most seven. Thus $(0, 11)$ is impossible.

For $(S, q) = (2, 11)$ the data are

$$(\mathrm{rk} G, \deg G, h^0(G)) = (7, 3, 8), \quad \deg M = -5, \quad h^1(M) = 8.$$

The equality types are

$$(1, 3) + (6, 0), \quad (2, 3) + (5, 0).$$

The rank-two, degree-three factor in the second type cannot have three sections, by the same exterior-square argument used at $q = 10$. Equality in the first type forces a splitting

$$G \simeq J \oplus \mathcal{O}_C^6,$$

where J is a basepoint-free trigonal pencil. The evaluation sequence is

$$0 \longrightarrow J^{-1} \longrightarrow \mathcal{O}_C^8 \longrightarrow G \longrightarrow 0.$$

The weight $S = 2$ comes from a unique type-2 + 1 marking p . Since $\det E \simeq \mathcal{O}_C(-2p)$ and $\det G = J$, one has

$$L = M^{-1} = J(2p).$$

The boundary $e : H^0(G) \rightarrow H^1(M)$ is an isomorphism. Project the quotient bracket to the $J \otimes L$ target and call its coefficient map ϕ . Čech symmetry gives an injection

$$H^0(\omega_C \otimes J^{-1}) \longrightarrow \mathrm{Sym}^2(\mathrm{im} \phi)^\vee$$

whose nonzero image forms are nondegenerate. The source has dimension two. If $\dim \mathrm{im} \phi = 1$, the target is only one-dimensional; if $\dim \mathrm{im} \phi \geq 2$, the image contains a complex pencil and therefore a singular form. Both are impossible, so $\phi = 0$. Since G is globally generated, the whole target- J bracket block vanishes.

Let $J' = \omega_C \otimes J^{-1}$ be the residual trigonal pencil. On a very general genus-four curve, every trigonal pencil is induced by one of the two rulings of the smooth canonical quadric: a trigonal pencil produces a ruling of the unique quadric containing the canonical curve. Each pencil has a finite ramification divisor. Since the marked point varies on the dominant pointed moduli cover, choose the test fibre so that p is unramified for every trigonal pencil, hence for J' . Then

$$h^0(J'(-2p)) = 0, \quad H^0(J(2p)) = H^0(J), \quad H^0(\mathcal{O}_C(2p)) = H^0(\mathcal{O}_C).$$

Both equalities of section spaces are induced by the natural inclusions. If z is a local parameter at p , then relative to the standard twisted frames $z^{-2}e_J$ of $J(2p)$ and z^{-2} of $\mathcal{O}_C(2p)$, these inclusions are multiplication by z^2 . After the target- J bracket block has vanished, every remaining matrix coefficient of β lies in one of these two images. Consequently every coefficient is divisible by z^2 , and its reduction on the contracted fibre is zero:

$$\beta_p = 0.$$

Saturation makes $M_p \neq 0$, so M_p would be a one-dimensional ideal in the contracted type-2 + 1 algebra

$$\mathfrak{gl}_2 \ltimes (\mathbb{C}^2 \oplus (\mathbb{C}^2)^\vee).$$

No such ideal exists: a line in the radical would be a one-dimensional $\mathfrak{gl}_2$ -submodule, while a line projecting nontrivially to $\mathfrak{gl}_2$ would have to project to its center, whose nontrivial action on the radical violates ideality. Thus (2, 11) is impossible.

6.9 The no-high $q = 12$ row is eliminated

The only row is $(S, q) = (2, 12)$, with

$$(\mathrm{rk} G, \deg G, h^0(G)) = (7, 4, 9).$$

The exhaustive equality types are

$$(1, 4) + (6, 0), \quad (2, 4) + (5, 0), \quad (3, 4) + (4, 0),$$

and in each case the degree-four factor of rank $r = 1, 2, 3$ would need $r + 2$ sections. The degree-four factor lemma rules out all three. Hence $(2, 12)$ is impossible. The $q = 12, 13$ zero-weight rows were already eliminated by the original finite section bounds.

Lemma 6.1 (Equality in the BPGN induction). *Let C be a nonhyperelliptic curve of genus $g \geq 2$, and let H be a semistable bundle satisfying*

$$0 \leq \mu(H) \leq 2g - 2, \quad h^0(H) = \mathrm{rk} H + \frac{\deg H}{2}.$$

Then $\deg H$ is divisible by $2g - 2$.

Proof. Follow the induction in the proof of [1, Theorem 2.1]. For its maximal-slope subbundle H_1 and semistable quotient H_2 , one has

$$h^0(H) \leq h^0(H_1) + h^0(H_2) \leq \mathrm{rk} H + \frac{\deg H}{2}.$$

Equality for H forces equality throughout, so both H_1 and H_2 again attain the Clifford bound. Iteration reduces to line bundles attaining classical Clifford equality. No nonspecial term can attain equality in the stated slope range: Riemann–Roch would force its slope to be $2g$. On a nonhyperelliptic curve, the only line bundles attaining classical Clifford equality are $\mathcal{O}_C$ and ω_C , of degrees zero and $2g - 2$. Additivity of degree along the induction filtration proves the claim. $\square$

6.10 The no-high $q = 14$ row is eliminated

For $(S, q) = (0, 14)$ one has

$$(\mathrm{rk} G, \deg G, h^0(G)) = (7, 8, 11).$$

The exact HN enumeration has only one type attaining the BPGN bound eleven, namely the single factor $(7, 8)$; every unstable type has bound at most ten. Thus G is semistable (indeed stable, since $\gcd(7, 8) = 1$) and equality holds in the vector-bundle Clifford inequality

$$h^0(G) = \mathrm{rk} G + \frac{\deg G}{2}.$$

Apply Lemma 6.1. In genus four an equality bundle has degree divisible by $2g - 2 = 6$, contradicting $\deg G = 8$. Hence $(0, 14)$ is impossible, and **the complete dense rank-eight genus-four branch is eliminated.**

6.11 The high-HN $q = 7$ layer is eliminated

The three genus-four high-HN rows with $q = 7$ are

$$(t, y) = (1, 5), (2, 11), (3, 17), \quad S \geq 2.$$

For all three, $B = A/N$ has rank $7 - t$ and degree $6(7 - t)$. Every HN slope of B is at most six, so equality of its mean slope forces B to be semistable of slope six. Put $H = B^\vee \otimes \omega_C$. Then H is semistable of degree zero. The exact sequence

$$0 \longrightarrow H \longrightarrow G \longrightarrow N^\vee \otimes \omega_C \longrightarrow 0$$

has a negative-slope right term, while $h^0(G) = 4$. Hence $h^0(H) = 4$. Successively saturating its nowhere-vanishing sections in the semistable degree-zero bundle gives a saturated evaluation subbundle

$$W = \mathcal{O}_C^4 \subset G.$$

Let $Q = G/W$. From

$$0 \longrightarrow H/W \longrightarrow Q \longrightarrow N^\vee \otimes \omega_C \longrightarrow 0$$

one obtains

$$\mathrm{rk} Q = 3, \quad \deg Q = 1 - S, \quad \mu_{\max}(Q) \leq 0.$$

Parabolic stability gives $H^0(E) = 0$, hence also $H^0(P) = 0$ for the inverse image P of W . Since $\deg M = -1$ and $h^1(M) = 4$, the inverse-image extension has boundary $e : H^0(W) \xrightarrow{\sim} H^1(M)$. Let

$$f : M \otimes W \longrightarrow Q, \quad \phi : H^0(W) \longrightarrow H^0(Q \otimes L)$$

be the quotient bracket and its coefficient map. The same Čech comparison gives $f(v)_*e(u) = f(u)_*e(v)$. If $V = \mathrm{im} \phi$ and $k = \dim V$, the resulting Petri map lands in $\mathrm{Sym}^2(V^\vee)$.

Let $I \subset Q$ be the saturated image of f , with rank r and degree d . Every HN slope of $I \otimes L$ is at most one. The small-slope section bound gives $h^0(I \otimes L) \leq r$. Since V generically spans $I \otimes L$, one has $r \leq k \leq r$, so $k = r$.

If $r = 2$, then $-2 \leq d \leq 0$. For the quotient line $R = Q/I$, Riemann–Roch gives

$$\mathrm{rk} \Theta = 6 - d + h^0(Q) - h^0(R) \geq 6.$$

Indeed $h^0(R) \leq 1$ because $\deg R \leq 1$; if it is one, then $\deg R \geq 0$ forces $d \leq -1$. This contradicts $\dim \mathrm{Sym}^2(\mathbb{C}^2) = 3$. If $r = 3$, the Petri map is injective, but

$$h^0(Q^\vee \otimes \omega_C) = 8 + S + h^0(Q) \geq 10 > 6 = \dim \mathrm{Sym}^2(\mathbb{C}^3).$$

Thus $r = k \leq 1$. Consequently $W_0 = \ker \phi$ has dimension at least three. If $h^0(L) = 0$, its remaining self-block vanishes. If $L \simeq \mathcal{O}_C(p)$, write that block as sA . Extension compatibility gives $s_*eA = 0$, while $s_* : H^1(M) \rightarrow H^1(\mathcal{O}_C)$ is an isomorphism by the exact sequence for $0 \rightarrow \mathcal{O}_C(-p) \rightarrow \mathcal{O}_C \rightarrow k(p) \rightarrow 0$. Hence $A = 0$. Therefore

$$\mathcal{O}_C^3 \subset \ker \beta.$$

Parabolic stability gives $H^0(E) = 0$, so the full boundary $H^0(G) \rightarrow H^1(M)$ is injective. The normalizer boundary-cocycle obstruction excludes the rank-three trivial kernel. This closes all three $q = 7$ rows. The argument uses closure of the canonical parahoric lattice under bracket, but no determinant effectivity, puncture-type exclusion, or relative-section theorem.

6.12 The high-HN $q = 8$ layer is eliminated

The four genus-four high-HN rows with $q = 8$ are

(t, y)	Weight condition	$(\text{rk } H, \deg H)$
$(1, 4)$	$S \geq 3$	$(6, 0)$
$(1, 5)$	$S \geq 2$	$(6, 1)$
$(2, 10)$	$S \geq 3$	$(5, 0)$
$(2, 11)$	$S \geq 2$	$(5, 1)$

Here

$$\deg M = -2, \quad \deg L = 2, \quad h^0(G) = h^1(M) = 5, \quad \deg G = 2 - S,$$

and $H = (A/N)^\vee \otimes \omega_C \subset G$. The quotient $N^\vee \otimes \omega_C$ has negative HN slopes, so $H^0(H) = H^0(G)$. We treat the degree-zero and degree-one evaluation types separately.

For $(t, y) = (1, 4), (2, 10)$, the bundle H is semistable of degree zero and its five sections give a saturated

$$W = \mathcal{O}_C^5 \subset G.$$

The quotient $Q = G/W$ has rank two, degree $2 - S$, and $\mu_{\max}(Q) \leq 0$. Thus $Q \otimes L$ has determinant of degree at most three and every line subbundle has degree at most two. If it had three independent sections, the wedge map to its determinant would have a decomposable kernel and would produce a degree-at-most-two pencil. Hence $h^0(Q \otimes L) \leq 2$. If a coefficient-rank-two bracket had saturated image a line I , then $\mu_{\max}(Q) \leq 0$ would give $\deg(I \otimes L) \leq 2$, producing the same forbidden pencil. Its image must therefore have full rank, so the Čech–Petri map is injective. But

$$h^0(Q^\vee \otimes \omega_C) = 4 + S + h^0(Q) \geq 7 > 3 = \dim \text{Sym}^2(\mathbb{C}^2).$$

The coefficient rank is therefore at most one. The degree-two self-block can lose at most one further dimension, so three of the five universal sections lie in the quotient normalizer. The normalizer boundary-cocycle obstruction eliminates both degree-zero rows.

For $(t, y) = (1, 5), (2, 11)$, the HN-piece section bounds force the saturated evaluation bundle to have rank five and degree zero or one. Degree zero gives $\mathcal{O}_C^5$. Degree one gives

$$W = J \oplus \mathcal{O}_C^4, \quad J \simeq \mathcal{O}_C(p), \quad F := \text{im}(\mathcal{O}_C^5 \rightarrow G), \quad W/F \simeq k(p).$$

When $\text{rk } H = 5$, as in $(2, 11)$, only the degree-one alternative is possible. The saturated $\mathcal{O}_C^5$ alternative occurs only in the row $(1, 5)$.

In that saturated alternative, Q has rank two, degree $2 - S$, and $\mu_{\max}(Q) \leq 1$. If the saturated bracket image is a line and has two coefficient sections, it has degree one before twisting, so its twist by L is a basepoint-free trigonal pencil A . For the quotient line $R = Q/I$, the actual restriction image

$$T = \text{im}(H^0(Q^\vee \otimes \omega_C) \longrightarrow H^0(I^\vee \otimes \omega_C))$$

has dimension $2 + h^0(Q) \geq 2$. Čech symmetry says that $\alpha H^0(A)$ is the same fixed two-plane for every nonzero $\alpha \in T$. Two independent choices would have a nonconstant ratio preserving the pencil; the eigenvector argument makes that ratio constant, a contradiction.

If the bracket image has full rank, coefficient rank two is excluded by

$$h^0(Q^\vee \otimes \omega_C) = 4 + S + h^0(Q) \geq 6 > 3.$$

For coefficient ranks greater than two, use

$$0 \longrightarrow (H/W) \otimes L \longrightarrow Q \otimes L \longrightarrow N^\vee \otimes \omega_C \otimes L \longrightarrow 0.$$

The two line terms have at most two and one sections, respectively, so $h^0(Q \otimes L) \leq 3$. Thus the only remaining full-image possibility is coefficient rank three. Its Petri source has dimension at least six. If the inequality is strict, it cannot inject into $\text{Sym}^2(\mathbb{C}^3)$; in the equality case the map would be onto, while the evaluation construction supplies a nonzero reducible conic $\text{sym}(u_x \otimes \ell_x)$ in its zero-dimensional orthogonal. Thus this case is impossible as well. The saturated row again has coefficient rank at most one and yields three normalizer sections after the one-dimensional self-block loss.

It remains to handle the degree-one saturation defect. Put $R_0 = G/W$. Then R_0 has rank two, degree $1 - S$, and $\mu_{\max}(R_0) \leq 0$. A rank-one residual bracket image has at most one coefficient section after twisting by L . A full image has at most two coefficient sections by the same wedge argument, while coefficient rank two would inject a space of dimension

$$h^0(R_0^\vee \otimes \omega_C) = 5 + S + h^0(R_0) \geq 7$$

into a three-dimensional symmetric square. Hence the residual coefficient rank is at most one.

Suppose it is one and write $\phi(v) = c(v)r$ with $0 \neq r : M \rightarrow R_0$. Čech symmetry gives $\dim \ker r_* \geq 4$. If I is the saturation of $r(M)$ and $d = \text{length}(I/r(M))$, then $0 \leq d \leq 2$. The quotient $Q_r = R_0/r(M)$ fits into

$$0 \longrightarrow I/r(M) \longrightarrow Q_r \longrightarrow R_0/I \longrightarrow 0, \quad \deg(R_0/I) = 3 - S - d.$$

For $d = 0, 1, 2$, the section bounds for R_0/I are $1, 1, 0$, so $h^0(Q_r) \leq 1, 2, 2$. The cohomology sequence therefore gives $\dim \ker r_* \leq 2$, a contradiction. Thus the residual coefficient map vanishes.

Finally, the length-one principal-part map $W/F \simeq k(p)$ loses at most one of the five row sections. If L is effective, extension compatibility and the degree-two divisor sequence lose at most one more. At least three generically independent sections remain in the quotient normalizer, which is impossible by the boundary-cocycle proposition. The canonical adjoint parahoric lattice is closed under bracket, so this calculation remains holomorphic even when the evaluation-defect point is a puncture.

This eliminates every high-HN $q = 8$ row. The next layer is $q = 9$, which is also eliminated below; the remaining genus-four high-HN rows have $q \geq 10$.

6.13 The high-HN $q = 9$ layer: an apparent self-block wall and its closure

The four high-HN rows at $q = 9$ are

(t, y)	Weight condition	$(\text{rk } H, \deg H)$
$(1, 3)$	$S \geq 4$	$(6, 0)$
$(1, 4)$	$S \geq 3$	$(6, 1)$
$(1, 5)$	$S \geq 2$	$(6, 2)$
$(2, 11)$	$S \geq 2$	$(5, 2)$

Here again $H = (A/N)^\vee \otimes \omega_C$, while

$$\deg M = -3, \quad \deg L = 3, \quad h^0(G) = h^1(M) = 6, \quad \deg G = 3 - S.$$

The rank-five row $(2, 11)$ is impossible. Every HN factor of H has nonnegative degree and the total degree is two. The small-slope bounds give at most its rank many sections for every degree-zero or degree-one factor and for every rank-at-least-two degree-two factor; a degree-two line has at most one section on the nonhyperelliptic fibre. Hence $h^0(H) \leq 5$, contrary to $h^0(H) = h^0(G) = 6$.

For the three $t = 1$ rows, put $\delta = \deg H = 0, 1, 2$. The six sections have full-rank raw evaluation

$$F \simeq \mathcal{O}_C^6 \longrightarrow H, \quad \text{length}(H/F) = \delta.$$

The residual quotient line $Q = G/H = N^\vee \otimes \omega_C$ has degree $3 - S - \delta$, so $Q \otimes L$ has degree at most two. Its residual bracket coefficient therefore has rank at most one. If it had rank one, write it as $\phi(v) = c(v)r$ with $0 \neq r : M \rightarrow Q$. Čech symmetry and the boundary isomorphism would give

$$\dim \ker(H^1(M) \xrightarrow{r_*} H^1(Q)) \geq 5.$$

But $Q/r(M)$ is torsion of length $6 - S - \delta \leq 2$, and $H^0(Q) = 0$. The cohomology sequence gives $\dim \ker r_* \leq 2$, a contradiction. Thus the residual coefficient vanishes in all three rows.

Only the principal parts and the degree-three self-block remain. Killing the δ principal parts leaves at least $6 - \delta$ raw sections. If $h^0(L) = 0$, the self-block vanishes and the resulting normalizer ranks are $6, 5, 4$; all three rows close. If $h^0(L) = 1$, multiplication $H^1(M) \rightarrow H^1(\mathcal{O}_C)$ has a two-dimensional kernel. The guaranteed normalizer ranks are then

$$(4, 3, 2) \quad \text{for } \delta = (0, 1, 2).$$

The first two rows close by this count. At this stage the row $(t, y) = (1, 5)$ retains a maximal-defect possibility in which both the principal-part map and the self-block have rank two.

If $h^0(L) = 2$, then L is a basepoint-free trigonal pencil. The combined compatibility map from a twelve-dimensional space to $H^1(\mathcal{O}_C)$ is surjective and has kernel dimension eight. As in the no-high $q = 9$ branch, that kernel admits an injective six-dimensional matrix pencil, so the extension equations alone force no constant normalizer kernel. The earliest such wall is

$$(t, y, q) = (1, 3, 9), \quad S \geq 4, \quad h^0(L) = 2.$$

At the level of the extension equations this is a genuine method wall, not a geometric adjoint counterexample: generic Kirillov, trace, Jacobi, and orbit-type identities alone are not enough. The canonical local lattice supplies the additional input that closes it below.

The sharp primitive pairing-ideal bound above removes every effective row, including the apparent defect-zero trigonal wall. The quadratic pushout first gives $\text{tr}(m^2) = 0$, and the symmetric Čech identity gives $P \subset M^\perp$. The global degree comparison then requires $\deg D_b \geq S$, whereas the canonical local basis gives $\deg D_b \leq S/2$. Since every high-HN $q = 9$ row has $S > 0$, this is a contradiction. Together with the already closed ineffective- L subloci and the impossible rank-five

row, **the entire genus-four high-HN $q = 9$ layer is impossible.** The remaining numerical rows are treated next.

6.14 All remaining high-HN rows are eliminated

After the preceding $q \leq 9$ eliminations, the high-HN ledger contains only

q	(t, y)	Weight condition
10	(1, 4)	$S \geq 3$
10	(1, 5)	$S \geq 2$
11	(1, 5)	$S \geq 2$

There are no high-HN rows with $q \geq 12$. These last rows have a common one-line contradiction. Dualizing the contraction sequence and the HN sequence gives

$$0 \longrightarrow M \longrightarrow E \longrightarrow A^\vee \otimes \omega_C \longrightarrow 0,$$

and

$$0 \longrightarrow (A/N)^\vee \otimes \omega_C \longrightarrow A^\vee \otimes \omega_C \longrightarrow Q := N^\vee \otimes \omega_C \longrightarrow 0.$$

Thus Q is a genuine quotient line of E , with no torsion correction, and

$$\deg Q = 6 - S - y.$$

Let a and b denote the numbers of type-2 + 1 and full-flag nonscalar markings. Then $S = 2a + 3b$. Scalar markings give quotient weight zero, while the induced quotient-line weight at every nonscalar marking lies in $[0, 1)$. If w_Q is the total quotient weight, then

$$w_Q < a + b, \quad \text{pardeg } Q < 6 - y - S + a + b = 6 - y - a - 2b.$$

For $y = 4$ the condition $S \geq 3$ implies $a + 2b \geq 2$, so $\text{pardeg } Q < 0$. For $y = 5$ the condition $S \geq 2$ implies $a + 2b \geq 1$, and again $\text{pardeg } Q < 0$. Parabolic stability of the degree-zero canonical adjoint bundle requires every proper quotient line to have strictly positive parabolic degree. Both possibilities are therefore impossible. Consequently **every genus-four high-HN row is eliminated.**

7 Proper projective-closure coefficients in genus four

The dense rank-eight calculation is not by itself enough. The projective closure alternatives in the companion manuscript [6] also contain

Projective closure	Adjoint constituents
$\text{PGL}_2 \simeq \text{SO}_3$	$U_5 \oplus U_3$
monomial with component quotient S_3	$U_6 \oplus U_2$
monomial with component quotient C_3	$U_3^+ \oplus U_3^- \oplus \chi \oplus \chi^{-1}$

The rank-three, rank-two, and rank-one constituents are already below the fixed-part threshold in genus four. It remains to exclude a rank-one fixed part for U_5 and U_6 .

7.1 The symmetric universal-extension calculation

Both remaining coefficients carry an invariant symmetric form

$$B : E \otimes E \longrightarrow \mathcal{O}_C$$

which is holomorphic on the canonical Deligne lattice and nondegenerate over the function field. Reciprocal nonzero parabolic weights pair with one zero of B at a marking; this creates no pole or hidden $\mathcal{O}_C(D)$ target.

Suppose an inverse image $P \subseteq E$ fits into

$$0 \longrightarrow M \longrightarrow P \longrightarrow V \otimes \mathcal{O}_C \longrightarrow 0$$

and its boundary $e : V \rightarrow H^1(M)$ is an isomorphism. Put $L = M^{-1}$. The restriction of the pairing is a morphism

$$Q_2 = B|_{M \otimes M} : M \longrightarrow L.$$

Pairing P against M extends Q_2 , so the pushout of the extension along Q_2 splits and

$$(Q_2)_* e = 0.$$

Whenever $H^0(\omega_C \otimes L^{-1}) \neq 0$, a nonzero Q_2 would have injective Serre-dual multiplication

$$H^0(\omega_C \otimes L^{-1}) \xrightarrow{Q_2} H^0(\omega_C \otimes L),$$

contradicting $(Q_2)_* = 0$ and surjectivity of e . Thus $Q_2 = 0$ in all cases below. Pairing with M then descends on the trivial quotient to a map $c : V \otimes \mathcal{O}_C \rightarrow L$. Comparing the pairing in local splittings of the extension gives the symmetric Čech identity

$$c(v)_* e(u) + c(u)_* e(v) = 0 \quad \text{in } H^1(\mathcal{O}_C).$$

This identity will force $c = 0$, or will identify the inverse image P with $M^\perp$.

7.2 The rank-five orthogonal coefficient

Here $d = 5 = g + 1$. The companion manuscript's first-super proposition [6] excludes every total weight $S \neq 1$. Its weight-one classification leaves exactly

$$0 \longrightarrow M \longrightarrow E_5 \longrightarrow \mathcal{O}_C^4 \longrightarrow 0, \quad \deg M = -1,$$

with $e : \mathbb{C}^4 \simeq H^1(M)$. The local U_5 weights are $0, \pm\alpha, \pm 2\alpha$; total one occurs at a projective involution, with canonical weights $(0, 0, 0, \frac{1}{2}, \frac{1}{2})$.

The symmetric calculation gives $Q_2 = 0$: Riemann–Roch gives

$$h^0(\omega_C \otimes L^{-1}) = 2 + h^0(L) \geq 2.$$

Since $\deg L = 1$, one has $h^0(L) \leq 1$. If it is zero, the descended map $c : \mathcal{O}_C^4 \rightarrow L$ vanishes. If $L \simeq \mathcal{O}_C(p)$ with section s , then

$$s_* : H^1(M) \longrightarrow H^1(\mathcal{O}_C)$$

is an isomorphism by $0 \rightarrow M \xrightarrow{s} \mathcal{O}_C \rightarrow k(p) \rightarrow 0$. The symmetric Čech identity therefore forces $c = 0$ again. Thus M lies in the generic radical of B , contradicting generic nondegeneracy. Hence the rank-five constituent has no rank-one fixed part.

7.3 Exact rank-six monomial arithmetic

For U_6 , the positive local total weights are

$$\frac{3}{2}, \quad 2, \quad \frac{5}{2}, \quad 3,$$

and the global total S is integral. If a rank-zero Hodge vector existed, put

$$F = E_6^\vee \otimes \omega_C, \quad \deg F = 36 + S, \quad h^0(F) = 18 + S,$$

and let $A \subset F$ be its rank-five contraction kernel. Write $q = \deg(F/A) = 6 - \deg M \geq 7$.

In the no-high case the exact inequalities are

$$S \leq \frac{24}{7}, \quad \max(7, 6 + S) \leq q \leq 10 - S.$$

HN-piece section counting gives

S	q	$\deg(E_6/M)$	$h^0(E_6/M)$	HN upper bound
0	7	1	4	4
0	8	2	5	4
0	9	3	6	4
0	10	4	7	5
2	8	0	5	5

Thus only $(S, q) = (0, 7)$ and $(2, 8)$ remain. In the high-HN case the exact offsets are

$$t = 1 : y = 5, 6, \quad t = 2 : y = 12,$$

where $y = \deg N - S$. At the top offsets $y = 6, 12$, the genuine quotient $Q = N^\vee \otimes \omega_C$ has ordinary degree $-S$. Its induced quotient weight is at most the total coefficient weight S , so $\text{pardeg } Q \leq 0$, contradicting stability. The sole strict row is

$$(t, y, q) = (1, 5, 7), \quad S \geq 2.$$

Here $Q = N^\vee \otimes \omega_C$ is a genuine quotient line of E_6 with $\deg Q = 1 - S$. If m is the number of nonscalar markings, the local total list gives

$$m \leq \left\lfloor \frac{2S}{3} \right\rfloor \leq S - 1.$$

The quotient weight is strictly below m , so

$$\text{pardeg } Q < 1 - S + m \leq 0,$$

contradicting parabolic stability. No high-HN row survives.

7.4 The two rank-six no-high rows

For $(S, q) = (2, 8)$, equality gives

$$E_6/M \simeq \mathcal{O}_C^5, \quad \deg M = -2, \quad \mathbb{C}^5 \simeq H^1(M).$$

The symmetric calculation gives $Q_2 = 0$, since $h^0(\omega_C \otimes L^{-1}) = 1 + h^0(L) > 0$. On the nonhyperelliptic test fibre $h^0(L) \leq 1$. If L is effective, multiplication $H^1(M) \rightarrow H^1(\mathcal{O}_C)$ has one-dimensional kernel, whereas a nonzero cross term would put a four-dimensional subspace into that kernel. Thus the cross term vanishes. Since the quotient is globally generated, M would be in the generic radical of B , a contradiction.

For $(S, q) = (0, 7)$, let $R \subset G = E_6/M$ be proper. Its inverse image in E_6 has degree $\deg R - 1$, so stability of the ordinary degree-zero coefficient gives $\deg R \leq 0$. Hence G is stable of rank five and degree one. The saturation of its four-section evaluation is generically generated and has degree zero; all four sections force its rank to be four. It is therefore

$$W = \mathcal{O}_C^4 \subset G.$$

Let $P \subset E_6$ be its inverse image. Then $H^0(P) = 0$ and the boundary $\mathbb{C}^4 \simeq H^1(M)$ is an isomorphism. The symmetric calculation gives $Q_2 = 0$ and kills the cross term on W , so $P \subseteq M^\perp$. Both bundles have rank five and degree minus one; hence

$$P = M^\perp, \quad M^\perp/M \simeq \mathcal{O}_C^4, \quad \det E_6 \simeq \mathcal{O}_C.$$

This contradicts the determinant of the actual S_3 monomial constituent. The diagonal torus factors on the six ordered roots multiply to one, while a transposition permutes them by three swaps. Consequently

$$\det U_6 = \text{sgn} : S_3 \longrightarrow \{\pm 1\}.$$

The component quotient is S_3 , so density makes this a nontrivial unitary determinant character. Its unpunctured holomorphic line cannot be trivial. This closes the last proper-closure coefficient.

8 Genus-four propagation

The preceding fixed-part vanishing holds on every connected finite-etale cover used in the companion manuscript [6]. Projective finite-index descent and direct Artin devissage have no further genus restriction. The reducible nonscalar multiplicity comparisons remain strict at $g = 4$:

$$(u, w) = (1, \leq 4), (2, \leq 3), (3, 1), \quad w < 2g - 2u = 6, 4, 2.$$

Lemma 8.1 (The standard subsystem on a finite-index base). *Let $B \rightarrow \mathcal{M}_{g,n}$ be a connected base whose mapping-class-group image has finite index, with labeled marked sections, and put*

$$H = R^1\pi_*\mathbb{C} \subset R^1\pi_*^\circ\mathbb{C}.$$

Then H is irreducible with Zariski-dense Sp_{2g} -monodromy, and the quotient $R^1\pi_^\circ\mathbb{C}/H$ is constant. Consequently every irreducible rank- $2g$ sub-local system of $R^1\pi_*^\circ\mathbb{C}$ equals H , and H has no nonzero invariant symmetric bilinear form.*

Proof. The residue sequence identifies the quotient with the constant local system

$$\ker\left(\mathbb{C}^n \xrightarrow{\Sigma} \mathbb{C}\right)(-1).$$

The standard action of $\text{Mod}_{g,n}$ on H has image $\text{Sp}_{2g}(\mathbb{Z})$. The image of a finite-index subgroup is again finite index, hence Zariski dense in Sp_{2g} ; the standard representation is therefore irreducible.

Let I be an irreducible rank- $2g$ sub-local system of the punctured cohomology. If $I \cap H = 0$, then I injects into the constant residue quotient. Its monodromy is then trivial, contradicting irreducibility and rank $2g > 1$. Thus $I \cap H \neq 0$; irreducibility and equality of ranks give $I = H$. Finally, the standard Sp_{2g} -representation preserves its alternating form and has no nonzero invariant symmetric form. $\square$

The scalar residual case needs one new equality argument. The companion manuscript's exact trivial-coefficient gap [6] says every nonzero sub-local system of $R^1\pi_*\mathbb{C}$ has rank at least $2g = 8$. A first nonzero scalar Artin invariant has image I of rank at most eight, hence exactly eight and irreducible. By Lemma 8.1,

$$I = R^1\pi_*\mathbb{C}, \quad (W^0)^\vee \simeq R^1\pi_*\mathbb{C}.$$

The left side is an adjoint $\mathfrak{sl}_3$ local system and therefore preserves its nondegenerate symmetric Killing form. The right side has Zariski-dense Sp_8 -monodromy and admits no invariant symmetric form. This contradiction closes the scalar equality case.

The remaining propagation margins are still strict. The isomonodromy range is $3 < 2\sqrt{5}$, and for a characteristic extension of total rank at most three,

$$n_id_i \leq 2 < 2g - 2d_i, \quad d_i \leq 2,$$

because the right side is at least four. The projective/linear integrality, unitary-conjugate, and socle-induction arguments therefore carry over with the same cited inputs.

Completion of Candidate theorem 1.1. A reducible semisimple rank-three representation has constituents of rank at most two, hence lies in the published strict range. For an irreducible system with infinite projective image, the imported projective-closure classification leaves the four rows in section 2. All constituents of rank at most three are already below the fixed-part threshold. The dense coefficient U_8 is eliminated by the no-high and high-HN arguments above; the proper-closure coefficients U_5 and U_6 are eliminated in section 7. Thus the required finite-cover adjoint fixed-part vanishing holds for every actual irreducible coefficient.

Projective finite-index descent and direct Artin devissage import that vanishing to the dominant versal base. Reducible nonscalar residual systems retain strict multiplicity margins. The only scalar equality is excluded in section 8 by the incompatibility between an adjoint symmetric Killing form and the standard Sp_8 representation. The remaining rigidity, integrality, unitary-conjugate, and semisimple nonabelian-Hodge steps are B5. Finally, for a characteristic extension of total rank at most three, $n_id_i \leq 2 < 2g - 2d_i$ for $d_i \leq 2$ at $g = 4$; the imported socle induction therefore handles the nonsemisimple case. This proves the stated conclusion conditional on B1–B5. $\square$

9 Status and remaining verification boundary

The status of Candidate theorem 1.1 is **CANDIDATE**. More precisely:

- The finite HN ledgers and the displayed dimension, determinant, and fibrewise Lie-algebra calculations are exact finite calculations recorded in this manuscript.
- The dense U_8 elimination is a proposed proof conditional on the fixed-part, canonical-lattice, parabolic, and extension interfaces B1–B4. The most sensitive new steps are the Čech–Petri symmetry, saturation of raw evaluation lattices, the Spin lift and parity comparison, the primitive pairing-ideal bound at punctures, and the equality analysis in the BPGN Clifford induction.
- The U_5 and U_6 eliminations additionally depend on the companion first-super classification and on the invariant symmetric form extending holomorphically on the canonical Deligne lattice.
- Formal propagation at scalar residual type uses the asserted Zariski-dense Sp_8 monodromy on every finite-index mapping-class base and the absence of an invariant symmetric form on its standard representation.
- The conclusion also relies on B5: rigidity, integrality, isomonodromy, finite-cover descent, and socle induction. Those steps are not reproved here.

Independent specialist reconstruction of these geometric steps remains necessary. Finite scripts check only enumerations and selected linear algebra; the TeX build checks only the document. Non-load-bearing countermodels and refuted side routes are intentionally omitted so that the shortest proof graph remains visible. No external referee acceptance, priority claim, or established theorem status is asserted.

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